

MATH-517: Assignment 3

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Theoretical exercise

Your answer goes here

Practical exercise

The goal of this simulation study is to see how changing parameters such as sample size (n), number of blocks (N), and the α and β values for the underlying beta distribution for the covariate X , impacts the optimal bandwidth h_{AMISE} . We do this for a regression function $Y = m(X) + \epsilon$ where $m(X) = \sin((\frac{x}{3} + 0.1)^{-1})$ and $\epsilon \sim N(0, \sigma^2)$. I set $\sigma^2 = 1$ for simplicity. I run a simulation 200 times, iterating through all the parameters (running through all combinations of α, β, n, N), and plot the average CP Mallow's and h_{AMISE} for each of the combinations.

From the graphs, we can see that as the number of blocks N increases, the optimal bandwidth h_{AMISE} decreases. In general, it decreases sharply at first but begins to stabilize as N continues to increase. This is because when there are fewer observations per block, a smaller bandwidth h could result in overfitting. This is also why as sample size n increases, h_{AMISE} decreases - because more observations are available per block, we can use a smaller bandwidth without overfitting (note: focus on comparing lines where $\alpha = \beta$ to reduce the impact of the Beta distribution parameters discussed below).

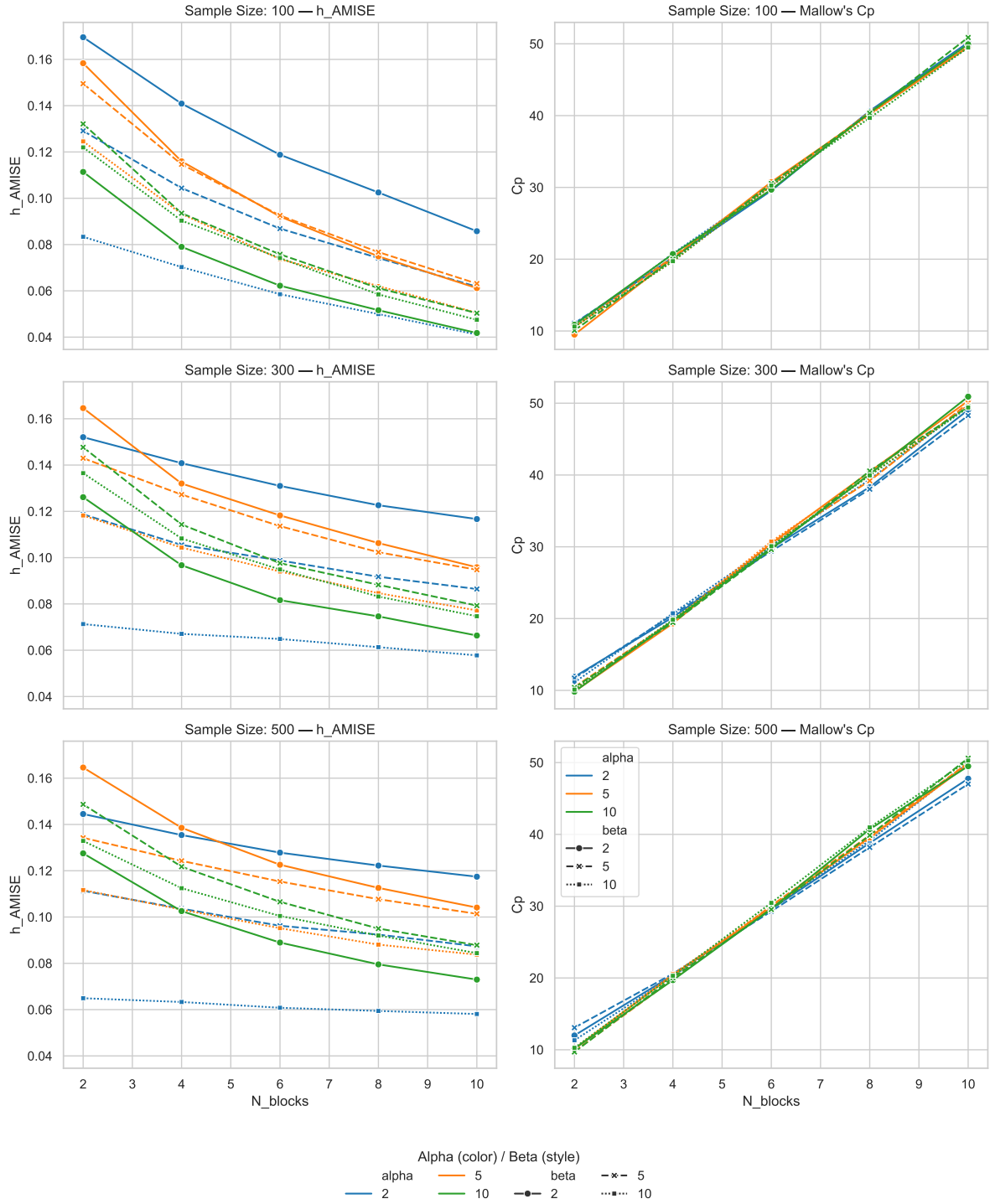
N should depend on n because it is important to have enough observations for each block. If n is too small and we choose too large of a N then each block N will have too few observations, and there will not be enough data to average out the impact of noise per block. There may be a risk of overfitting as the polynomial may be “wigglier” and overfit to noise. When n is large, more blocks can be chosen as more observations are available to fit data on within each block. There are many methods to determine an optimal N . One option is to find the value that minimizes Mallow's $C_p(N) = \frac{RSS(N)}{\frac{RSS(N_{\max})}{n - 5N_{\max}}} - (n - 10N)$ where

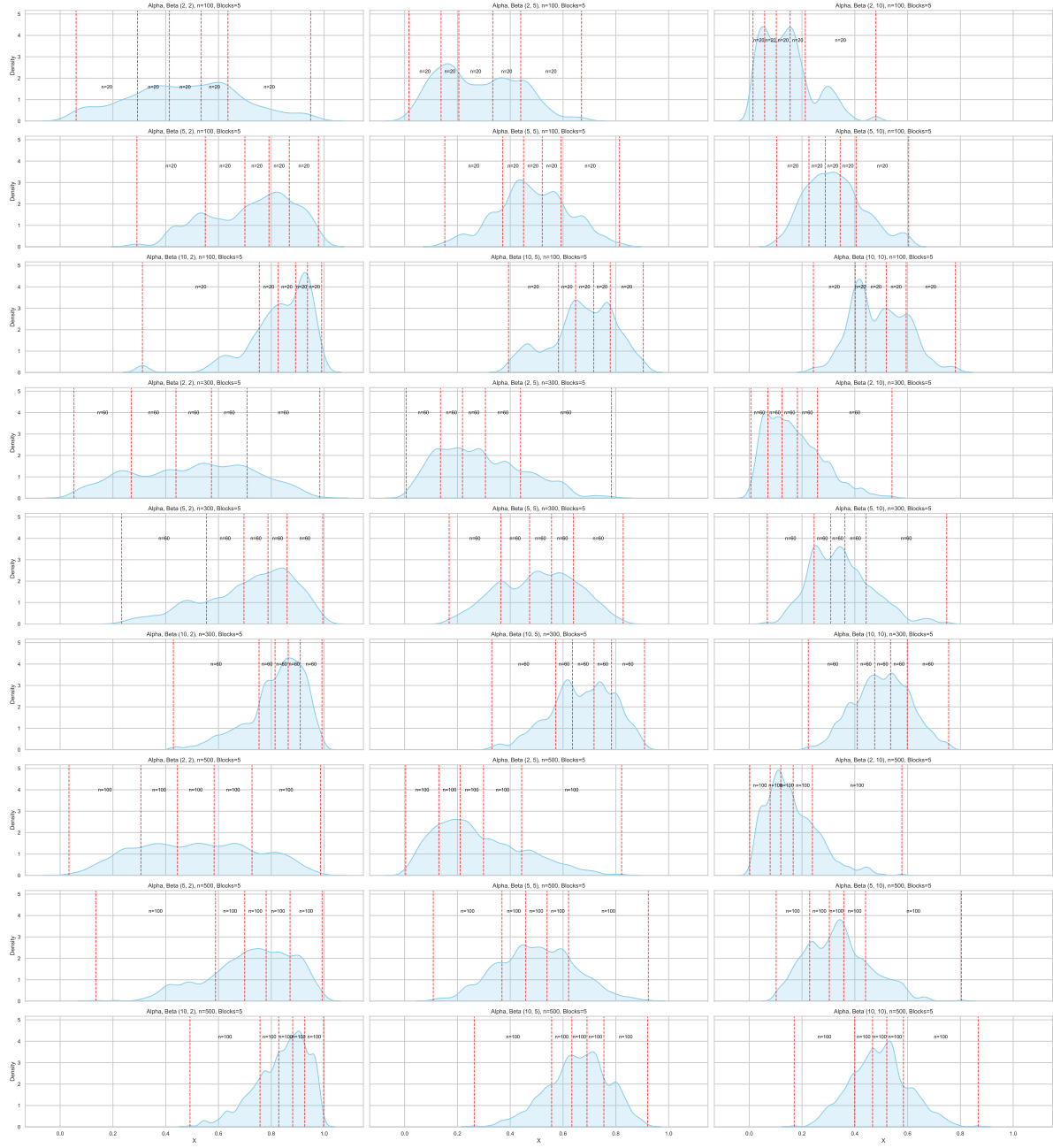
$RSS(N) = \sum_{i=1}^n \sum_{j=1}^N \{Y_i - \hat{m}_j(X_i)\}^2 \mathbf{1}_{X_i \in X}$ and $N_{\max} = \max\{\min(\lfloor \frac{n}{20} \rfloor, 5), 1\}$. In this simulation, Mallow's C_p is minimized for the minimum number of blocks ($N=2$).

The Beta distribution parameters (α, β) controls the shape of the distribution for X . As seen in the second set of graphs below, when α and β are significantly different, then the data is skewed. This means that observations cluster in some regions, but not in others - resulting in some blocks having fewer observations and reducing the reliability of the polynomials from those blocks.

I have also included a Shiny App in another file (`shiny_app.py`) to experiment with the parameters of the simulation. This can be run from the terminal (`shiny run shiny_app.py`).

Effect of Block Size N on h_{AMISE} and Mallow's C_p





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